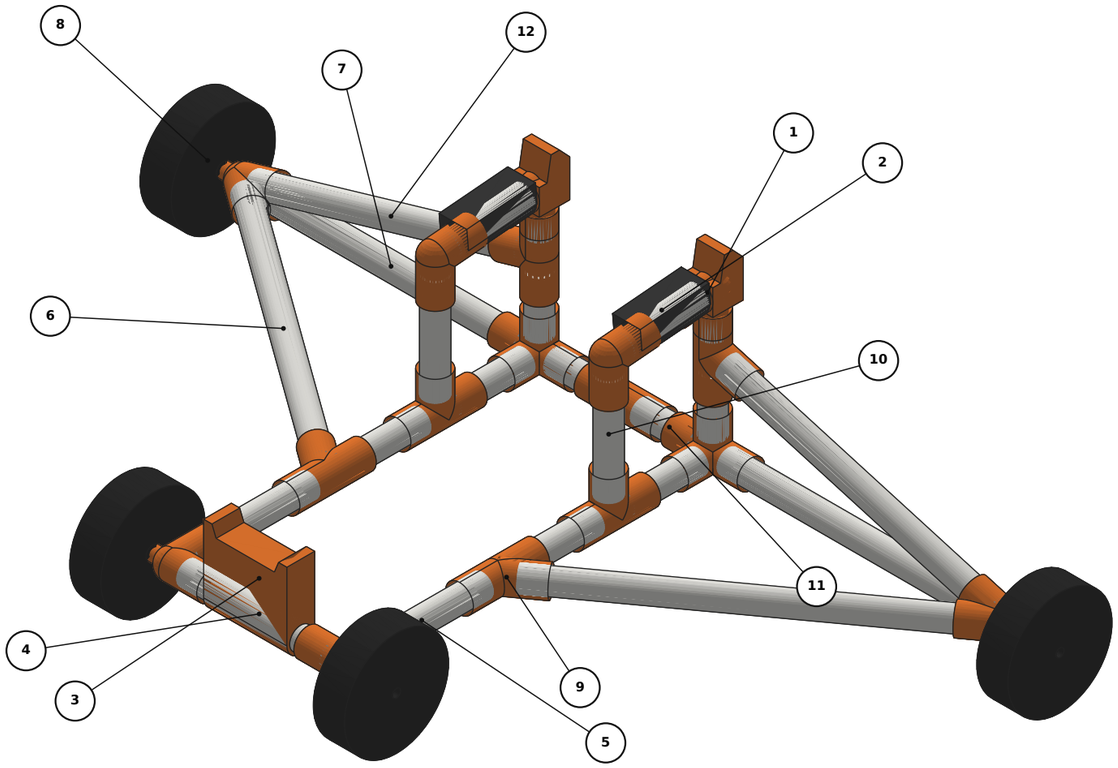
DOLLY - ISOMETRIC (REAR, STARBOARD, ABOVE)

aircraft removed for clarity



ITEMS

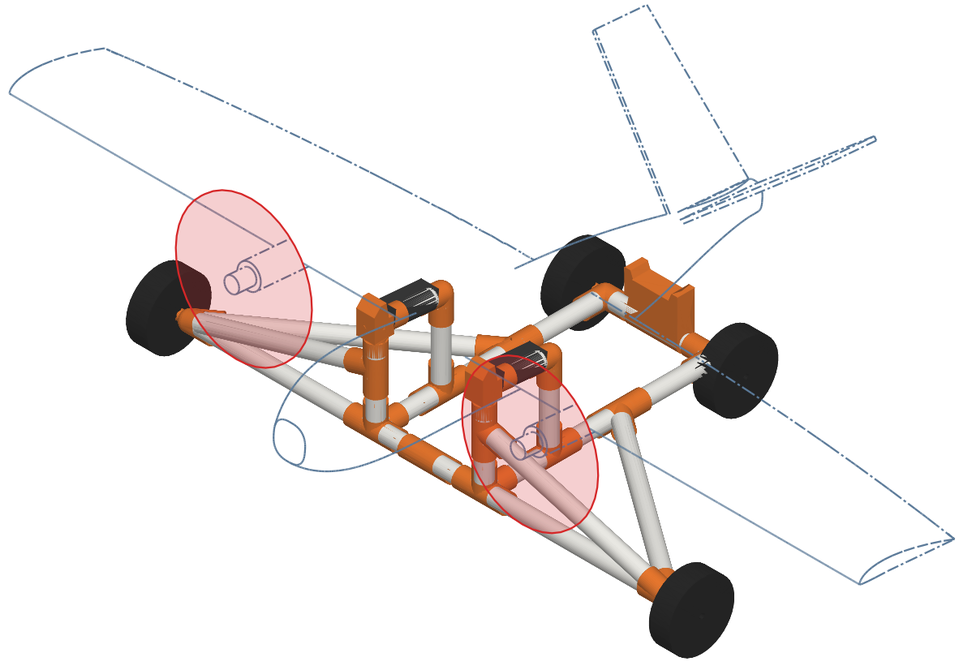
1. Ramp head (PETG, TPU face) - front stop; wing must lift to clear lip
2. TPU wing saddle pad on saddle tube, inboard of motor
3. Tail cradle (PETG, TPU pad) on rear cross member
4. Rear cross member (tail support)
5. Rail (each side, under the pads)
6. Outrigger brace
7. Front outrigger - wheels outside prop disc
8. Wheel 150 mm, grass tyre (x4)
9. Printed fittings (orange), screwed to tube
10. Rear upright (front upright under ramp head)
11. Front cross member (passes under fuselage)
12. Outrigger strut (vertical truss) to front upright at 200 mm

NOTES

1. Concept: the aircraft sits on two wing saddles and a tail cradle. Thrust pushes the wing leading edge against the ramp heads, so the aircraft pushes the dolly down the runway.
2. The aircraft cannot leave until wing lift carries most of its weight: it then pivots nose-up on the tail cradle and rides over the ramp lip. Lift-off speed is set by the resting wing angle.
3. The tail sits on the cradle, so elevator cannot rotate the aircraft on the dolly; rotation comes from wing lift.
4. All dolly structure stays clear of the 12 in prop discs; front wheels sit outboard of the discs.

ASSEMBLY WITH BELIEVER AT REST

aircraft envelope = phantom; red = propeller discs (12 in)



Min. prop-disc to dolly: 79 mm at rest, 75 mm at release (req. >= 75)

KEY DATA	VALUE
Wing support	2 x TPU saddle at y = +/-125, inboard of motors
Resting wing angle	5.0 deg (underside to ground)
Release angle	7.5 deg (stall approx 11.5 deg)
Ramp lip above wing LE	22.5 mm, face 80 deg
Est. lift-off speed	12.4 m/s at 4.0 kg
Est. ground roll	32 m / 4.9 s (Crr 0.20)
Motor shaft height	320 mm (REQ-DOL-11 >= 150)
Overall L x W	679 x 1251 mm
Est. dolly mass	3.3 kg (4.5 m of 32 mm PVC)

Airframe dims marked ASSUMED are not yet measured - verify before cutting tube.

QUT AEROSPACE SOCIETY - BELIEVER UAV — **CONCEPT**

LAUNCH DOLLY		DWG No.
CONCEPT ISOMETRIC		**BLV-DOL-001**
SCALE NTS (A3)	SHEET 1 OF 1	REV B 2026-11-04
UNITS mm | ANGLES deg	GENERATED: dolly_drawings.py	PROJECTION